
Development of a Test and Practice Platform for a Time Perception Research

Cristian Páez Rincón

Universidad El Bosque
cpaezr@unbosque.edu.co

Isaac Montoya Guzmán

Universidad El Bosque
jimontoya@unbosque.edu.co

Carlos López Ospina

lopezcarlosa@unbosque.edu.co

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the Owner/Author.
Copyright is held by the owner/author(s).
CSCW '17 Companion, February 25 - March 01, 2017, Portland, OR, USA
ACM 978-1-4503-4688-7/17/02.
<http://dx.doi.org/10.1145/3022198.3026361>

Abstract

UPDATED—29 December 2016. The Attention deficit hyperactivity is a behavior disorder with high global research. Because of this, his relation with the Time Perception is being research by the Neuroscience Institute of the Universidad el Bosque in order to find new ways of treatment of the Disorder, and the Institute needs new technological tools that help them finish their research. Seeing that, we have developed an activities manager software for the usage of the institute users and activities. Thanks to this system we have increase the coverage of the institute's research, reduced the costs of the patients and improve the reliability of the institute's research; providing a Time Perception activities and users manager for research propose.

Author Keywords

ADHD; Neuroscience; User management; Web-based test; LEAN.

Introduction

The neuroscience Institute of El Bosque University is studying the relation between time perception and attention deficit hyperactivity disorder (ADHD) [1]. According to the institute, if this relation exists, the time perception could be use as treatment for the ADHD patients. The institute developed tests and practices for measure the time perception on patients, looking for collect data to the study. The tests and practices are realized in the institute and require a doctor and the

patient. They use a chronometer to record the time in some tests and practices and are limited to the Power Point tools and functionalities. The institute is trying to improve the investigation increasing the volume of data that they can collect but exists two obstacles, the costs and coverage. The high costs appears because they need a doctor to collect the tests and practices data results. It means pay the hours that the doctor invested for apply the tests and practices. Coverage is the other issue into the institute investigation. Only the patients in Bogotá and with previous date, will be able to present the tests and practices. Patients require time and transport to go to the institute. They expend money and sometimes is hard to schedule a date. The use of technology, in the actuality is a tool that let us optimize and organize process in way that we can reach better results [2]. In the context presented by the institute, it intends use the technology like tool for the research [3]. Taking advantage the internet tendency, we were able to extend the coverage in the investigation of the institute for their studies. Developing a web application we were able to provide the institute with a tool for their investigation that let them improve the coverage and decrease the costs, by this way increasing the investigation data for obtaining better results in less time and generating less costs.

Solution

Seeing this problem, we proposed a web-based system called SESPET, that handles tests and practices for the institute, where the patients can access to the system from everywhere with an internet connection and the Institute can keep a reliable record of the tests being presented.

The SESPET system provide the tests and practices to the patients inscribed in the institute. When a patient log in the system this will provide the tests and practices assigned and in the specific order required by his

researcher. This was important because some of the target patients will have ADHD and the system should keep them focused in the tests and practices.

The scalability of the system was also an important issue for the institute, since the tests tend to change during the course of the research. For this we made the tests in a way that all his parameters can be changed and modified. Also we made that the system could be able to receive the code of new tests and practices so new tests could be added in a future.

For the system's access we created three user profiles: Administrator, Researcher and Patient. This user profiles have different access permissions to the system. The Patient can only perform tests and practices, the Researcher can assign those tests and practices to the patient with different values and see their results, and the Administrator can manage users, assign tests and practices, create new tests and practices and view of the tests results.

Finally, for the result's view we create a simple view that find a range of the significant results, since this are usually in milliseconds, and provide an export to excel functionality for an easy data analysis.

Construction

To develop the SESPET platform, we used Lean's development methodology combined with a Kanban board to take control of the tasks and generate an agile and quality process [4]. Inside the objectives and development there was an important selection phase of technology where were concluded the necessary tools needed for its development. Following Lean, the construction of the platform was made, looking forward to keep a high interaction with the client and generating just the necessary gadgets. About the management of the information, we used relational database MySQL because it allow us to make a business process modeling

(BPM) based on the institute needs and keeping low costs thanks to the free license. To use internet and keeping in mind the experience of developers, we used Java JEE to build the web platform. Inside this, we apply the model view controller architecture letting us to delegate the appropriate task to each responsible and separate the modules. To build the view we used PrimeFaces framework which allow us to be fast with the creation of the interface and also to make it attractive for users. With Hibernate [5], we map out all the database information to have a fast access to them from Java. Finally, we used spring [6] as a part of the controllers to manage the information exchange between the client and the server, it also allow us to provide a security standard to take control over who's allowed to access to a specific functionality of the platform. Keeping in mind this is a test and practice platform, was developed in the way that the institute can parameterize it. This mean the institute could, if they require, add new activities into the platform and can assign the required number of in and out variables to them. The activities are designed to can be develop using Javascript and HTML, bringing a high variety of possibilities to create new activities.

Results

The main reason of this project was the increase the research scope, reduce costs of the research and provide reliability in the times gathered by the tests. Therefore we run surveys with healthy and with different diseases looking for signs of this improvement.

Given the increase of the use of internet and computer's in Colombia, by implementing an internet platform that handles the tests and practices for the Institute we already made an improvement in scope. However, we use the surveys to support this fact asking people of all the regions of Colombia if was possible the access to the platform and if was possible to make the tests and

practices, we even include some test subjects in Mexico and the United States to support even further this functionality. We obtain that all the users were able to perform the tests and that their information was stored successfully when needed.

Referring to the costs of the Institute's research, we found that a cost for each appointment was between 60.000 and 80.000 COP (between 20 and 27 USD) for the institute and the costs of transportation for the patients was between 4.000 and 8.000 COP (between 1.35 and 2 USD), making it less appealing for the patients to keep with the support of the research. We tried to remove this costs using the SESPET platform, assuming that the use of an online test platform will reduce significantly the costs for appointments and transport for the patients. We supported this results by applying surveys to different users, we asked for the average costs of helping with the research of the Institute through an online platform and most of them say that was nearly nothing.

We also applied surveys to support the improvement in the user experience. We ask the researchers in the Institute about the experience using the platform and the improvements against the previous methods, obtaining clearly a better performance in the administration of users and tests for the Institute.

Conclusion

As a first conclusion we can affirm that the use of technological tools for research of the time perception is fundamental, since this provides a reliability in the data gathered from the patients, resulting in a better analysis of the study case and, therefore, better results of the researches.

Is important for the Institute that the ability to access the tests is available for most of the potential patients,

since this results in more patients, more data for the institute to analyze and best supported results. Therefore we can conclude that the increase of scope is an important improvement for the research of the institute. We can also conclude that the implementation of technological tools in Colombia is an important step, since the technological infrastructure is increasing every day.

We cannot talk about improvements in scope without talking about cost reduction, since the fact that the platform is on internet and available to all users makes the costs of the Institute and the patients go down by a significant amount, reducing the cost of appointments for the institute and deleting the costs of transportations for the patients. This brings more patients that are willing to help with the research and provides more data and support for the institute research. Finally, we can conclude that the use of technological tools, as the platform SESPET, for the development of researches of the time perception is fundamental, since the precision and reliability of the test have to be high in order to obtain meaningful results. Also, the implementation of technological tools for research institutes in Colombia provides better quality in Colombian researches, helping the country to improve the development of new knowledge.

References

- [1] Instituto de Neurociencias Universidad el Bosque. And Escuela Colombiana de Medicina. Percepción del tiempo; una visión global de las diferentes perspectivas en neurociencias.
- [2] S. G. Villate, M. E. Franzán, G. Passaglia, M. Cáceres and G. A .Ortiz. Trastornos en la percepción del tiempo en pacientes con alteraciones dopaminérgicas. *Neurología Argentina* 6(4), pp. 212-216. 2014.
- [3] J. Altés. Papel de las tecnologías de la información y la comunicación en lamedicina actual. *Seminarios De La Fundación Española De Reumatología* 14(2), pp. 31-35. 2013.
- [4] R. A. G. Bustos. Métodos de desarrollo de software: El desafío pendiente de la estandarización. *Software Development Methodologies: A Duel Pending for Standardization*.
- [5] Hibernate official site. - <http://hibernate.org/>
- [6] Spring official site - <https://spring.io/>